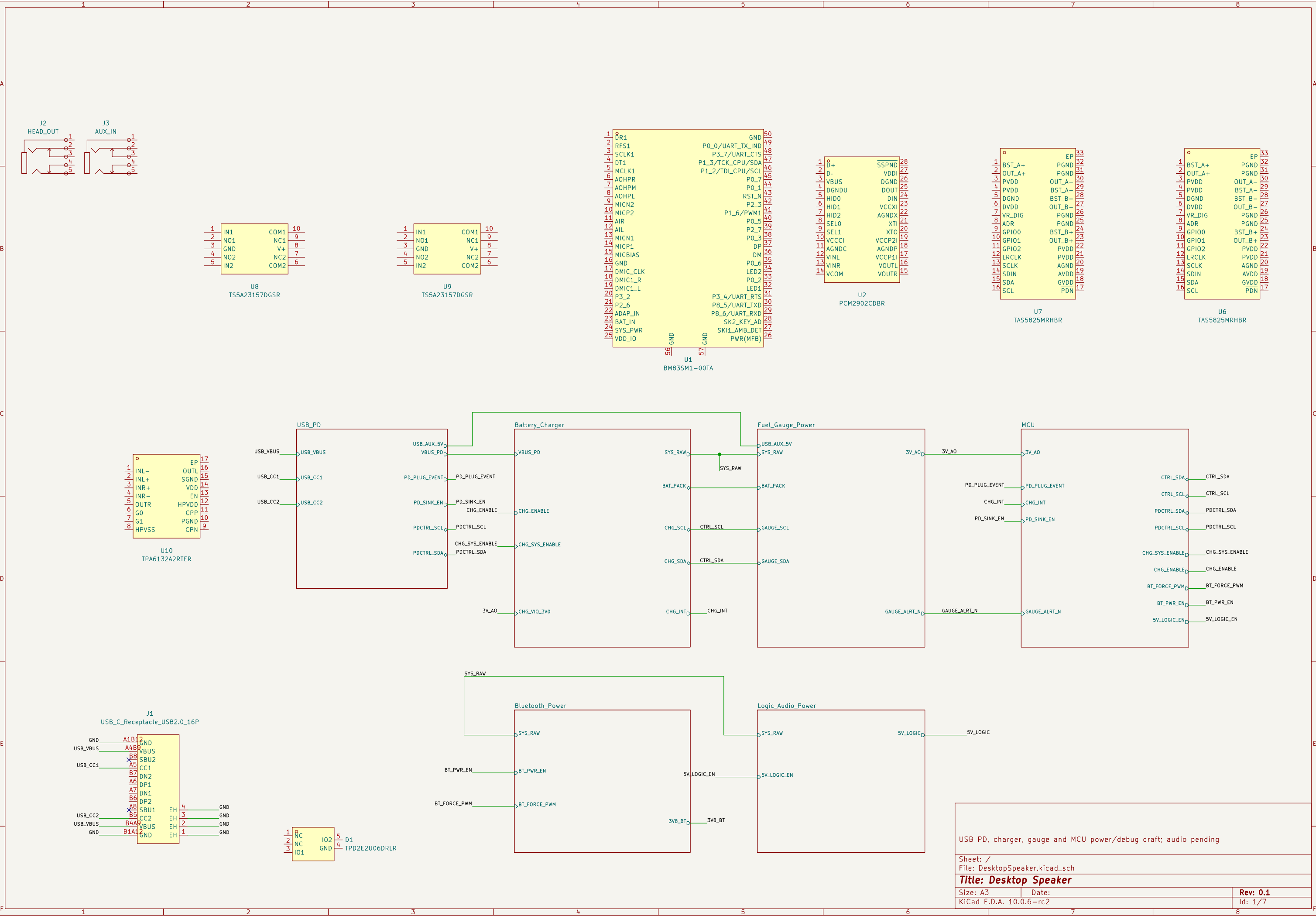
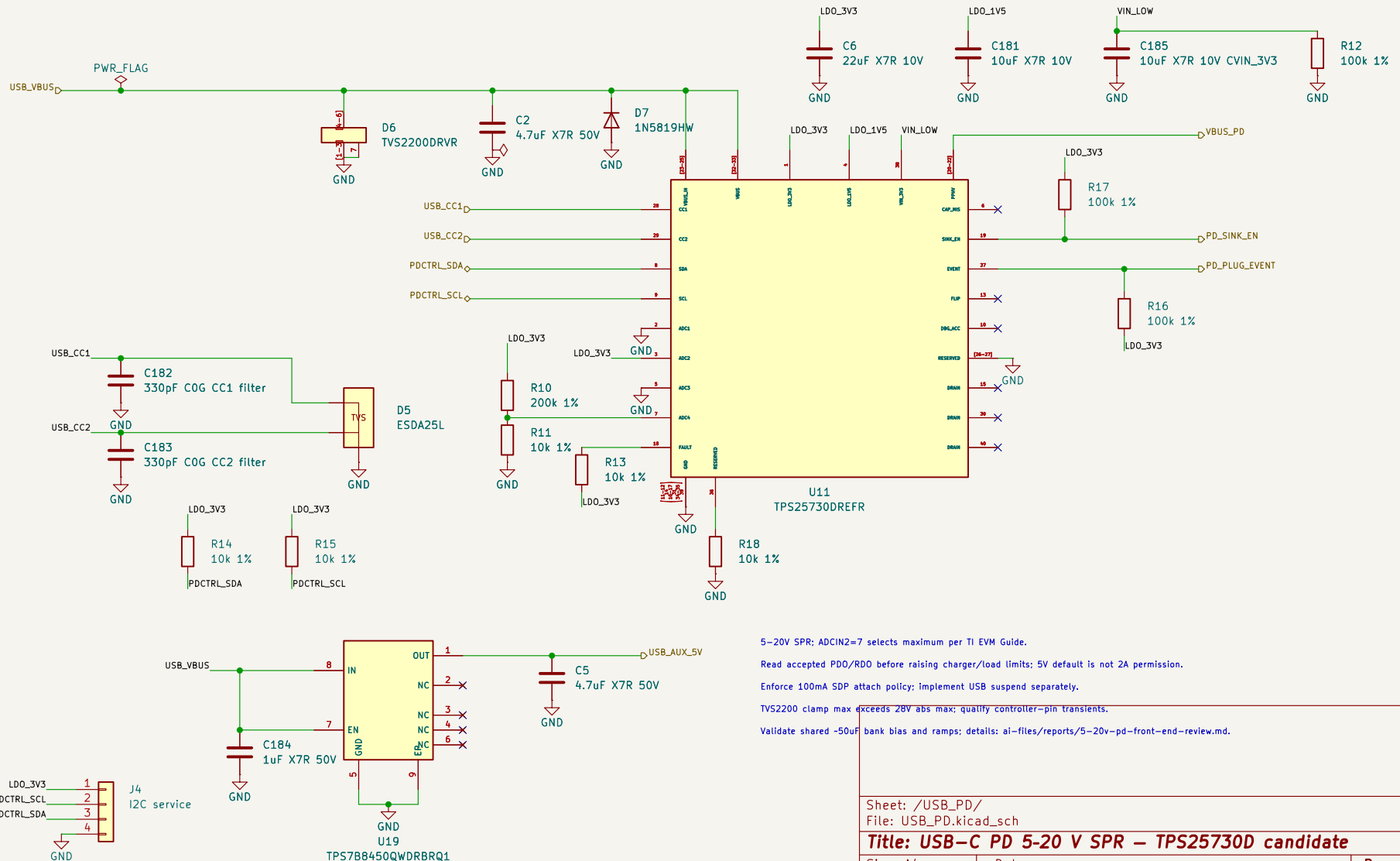






5-20V SPR; ADCIN2=7 selects maximum per TI EVM Guide.

Read accepted PDO/RDO before raising charger/load limits; 5V default is not 2A permission.

Enforce 100mA SDP attach policy; implement USB suspend separately.

TVS2200 clamp max exceeds 28V abs max; qualify controller-pin transients.

Validate shared -50uF bank bias and ramps; details: ai-files/reports/5-20v-pd-front-end-review.md.

Sheet: /USB_PD/

File: USB_PD.kicad_sch

Title: USB-C PD 5-20 V SPR - TPS25730D candidate

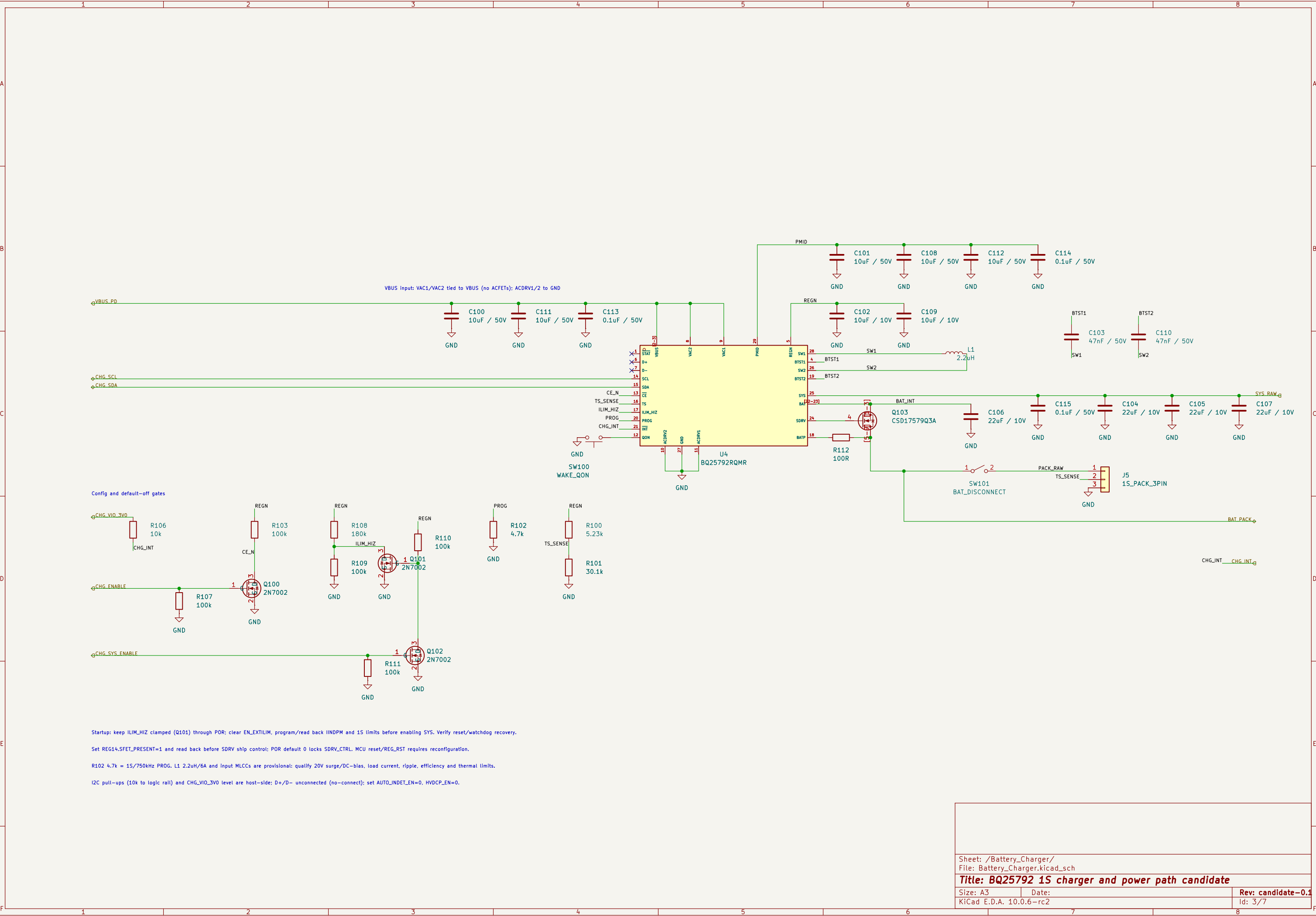
Size: A4

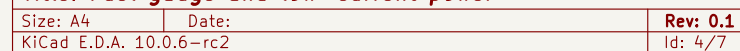
Date:

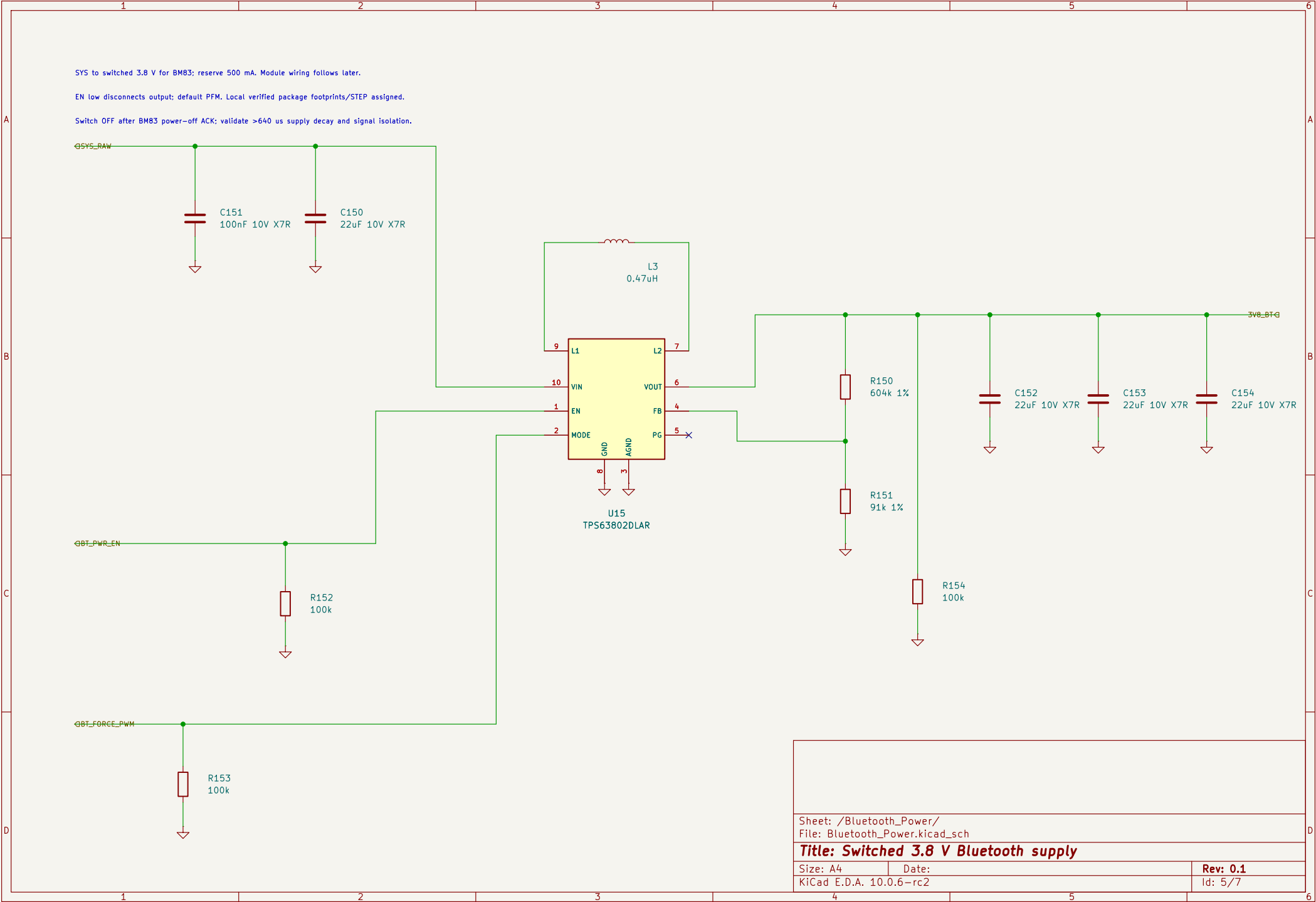
KiCad E.D.A. 10.0.6-rc2

Rev: prototype review

Id: 2/7



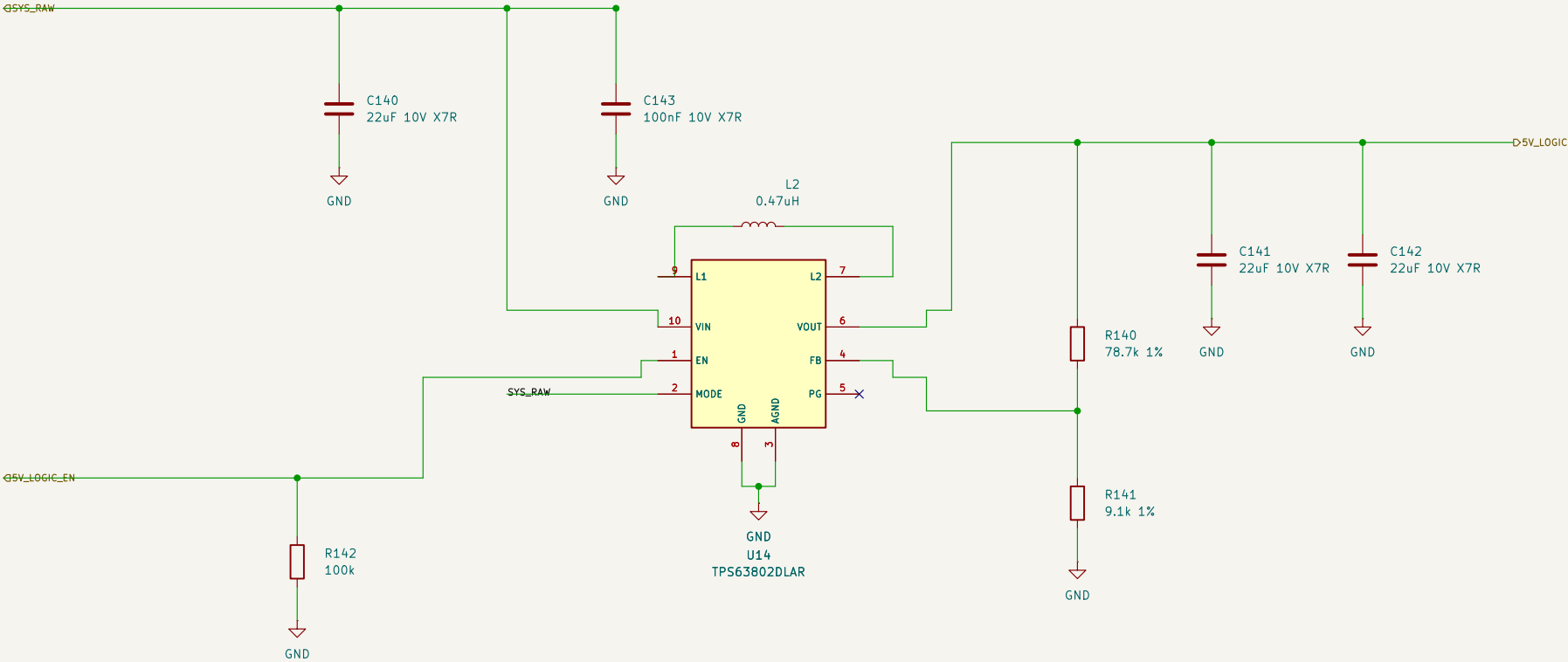




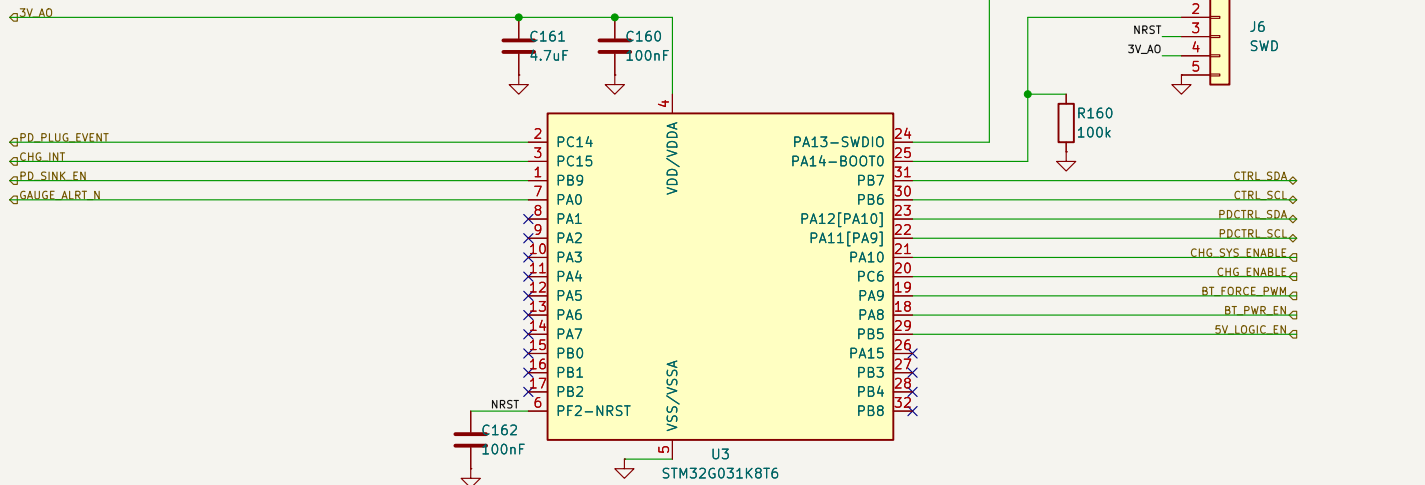
Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_A0 supply: 100nF + 4.7uF at VDD/VDDA (DS12992 fig. 13); NRST 100nF, Internal pull-up



Reserved (no-connect): see ai-files/reports/mcu-pin-allocation.md

STM32G031K8T6: supply, reset/boot, SWD and connections to captured power blocks

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU and programming

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 7/7